

Arjun Lakshmipathy

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Research Interests

Robotics: Manipulation, Soft Robotics, Motion Planning, Sensors, Locomotion
Graphics: Physics-Based Simulation & Modeling, Data Driven Animation, AR/VR, Discrete Differential Geometry
Optimization: Quadratic Programming, Convex and Non-Convex Methods, Numerical Approximation
Artificial Intelligence: Reinforcement Learning, Deep Learning, Heuristic Search, Statistical Inference
Imaging & Vision: Point Cloud Processing, Segmentation & Reconstruction Methods, Scene Understanding, Stereo Photography

Education

Carnegie Mellon University Ph.D. Computer Science Advisor: Nancy Pollard	GPA: N/A	Graduation: 2026
University of California, Los Angeles M.S. Computer Science Advisor: Demetri Terzopoulos	GPA: 3.814	Graduation: 2018
University of California, Los Angeles B.S. Computer Science & Engineering B.S. Mathematics & Economics	GPA: 3.668	Graduation: 2016

Research Appointments

Doctoral Research Assistant CMU Foam Robotics Lab	<i>September 2020 — Present</i>
Research Scientist NeuralX Incorporated	<i>March 2019 — August 2020</i>
Staff Research Associate UCLA Computer Graphics & Vision Lab	<i>January 2019</i>
Graduate Student Researcher UCLA Computer Graphics & Vision Lab	<i>November 2017 — December 2018</i>

Publications (Reverse Chronological)

Contact Transfer: A Direct, User-Driven Method for Human to Robot Transfer of Grasps and Manipulations
Arjun Lakshmipathy, Dominik Bauer, Cornelia Bauer, Nancy S. Pollard
Preprint - Submitted to IEEE International Conference on Robotics and Automation (ICRA'22)

Contact Tracing: A Low Cost Reconstruction Framework for Surface Contact Interpolation
Arjun Lakshmipathy, Dominik Bauer, Nancy S. Pollard
Proc. IEEE International Conference on Intelligent Robots and Systems (IROS'21), Prague, Czech Republic, September 2021

Locally-Connected, Irregular Deep Neural Networks for Biomimetic Active Vision in a Simulated Human
Masaki Nakada, Honglin Chen, **Arjun Lakshmipathy**, Demetri Terzopoulos
Proc. 25th International Conference on Pattern Recognition (ICPR'20), Milan, Italy, January 2021, 4465–4472

Deep Learning of Neuromuscular and Visuomotor Control of a Biomimetic Simulated Humanoid
Masaki Nakada, Tao Zhou, Honglin Chen, **Arjun Lakshmipathy**, Demetri Terzopoulos
IEEE Robotics and Automation Letters, 5(3), July 2020, 3952–3959. (Date of publication: February 10, 2020.)

Biomimetic Eye Modeling and Deep Neuromuscular Oculomotor Control
Masaki Nakada*, **Arjun Lakshmipathy***, Honglin Chen, Tao Zhou, Xin Ling, Demetri Terzopoulos
ACM Transactions on Graphics, 38(6), November 2019, 221:1–14. Proc. ACM SIGGRAPH Asia 19 Conference, Brisbane, Australia, November 2019

(* equal contributors)

Teaching Experience

Introduction to Computer Graphics (TA) | UCLA CS 174A

Fall 2017, Winter 2018

Basic principles behind modern two- and three-dimensional computer graphics systems, including complete set of steps that modern graphics pipelines use to create realistic images in real time. How to position and manipulate objects in scene using geometric and camera transformations. How to create final image using perspective and orthographic transformations. Basics of modeling primitives such as polygonal models and implicit and parametric surfaces. Basic ideas behind color spaces, illumination models, shading, and texture mapping.

Computer Networks: The Physical Layer (TA) | UCLA CS 117

Spring 2018, Summer 2018, Fall 2018

Introduction to fundamental computer communication concepts underlying and supporting modern networks, with focus on wireless communications and media access layers of network protocol stack. Systems include wireless LANs (IEEE802.11) and ad hoc wireless and personal area networks (e.g., Bluetooth, ZigBee). Experimental project based on mobile radio-equipped devices (smart phones, tablets, etc.) as sensor platforms for personal applications such as wireless health, positioning, and environment awareness, and experimental laboratory sessions included.

Technical Skills

C++ (Dynamic Animation and Robotics Toolkit, Point Cloud Library, OpenSceneGraph, Eigen, ImGUI, OpenGL + GLUT), Python (NumPy, Open3D, Tensorflow, PyTorch, SciPy), C, ROS, Blender, AWS (EC2, S3, EKS, ELB, ECS, CloudFront, Cognito, DynamoDB, MediaConvert, Route 53, ACM, SES, SNS), JavaScript (NodeJS / Express, ReactJS, AngularJS), Docker + Kubernetes, NGINX + Certbot, Jenkins, Java / Spring, Scala / Play, Elasticsearch, Matlab, Git

Graduate Coursework

Graduate Algorithms (CMU 15-750)	<i>Spring 2021</i>
Biomechanics and Motor Control (CMU 16-868)	<i>Fall 2020</i>
Computational Photography (CMU 15-862)	<i>Fall 2020</i>
Problem Solving and Search (UCLA CS 261A)	<i>Spring 2018</i>
Special Topics: Shared Economy Systems (UCLA CS 259)	<i>Spring 2018</i>
Artificial Life for Computer Graphics and Vision (UCLA CS 275)	<i>Winter 2018</i>
Neural Networks and Deep Learning (UCLA EE 239AS)	<i>Winter 2018</i>
Embedded Systems (UCLA CS M213A)	<i>Fall 2017</i>

Non-Research Professional Appointments*

Senior Tutor (Part Time)

March 2021 — Present

Cambridge Coaching

Technical Advisor (Part Time)

September 2020 — December 2020

NeuralX Incorporated

Founding Engineer

March 2019 — August 2020

NeuralX Incorporated

Full Stack Software Engineer

January 2017 — September 2017

Rally Health Incorporated

Software Engineering Intern

June 2015 — September 2015

Yahoo! Incorporated

Software Engineering Intern

June 2014 — September 2014

ViaSat Incorporated

iOS Development / Quality Control Intern

July 2013 — September 2013

Sandia National Laboratories

(*job titles only - full description and contributions available on LinkedIn)

Service

Reviewer, IEEE International Conference on Intelligent Robots and Systems (IROS) 2021 *2021*

UPE + TBP Free Tutoring *2013 — 2016*

Honors and Awards

Cum Laude Honors (BS) *June 2016*

Upsilon Pi Epsilon Honor Society Member *April 2013 — Present*

Tau Beta Pi Honor Society Member *April 2013 — Present*

References

1. **Nancy Pollard**
Professor
Computer Science Department, Robotics Institute
Carnegie Mellon University
npollard@andrew.cmu.edu
2. **Demetri Terzopoulos**
Distinguished Professor and Chancellor's Professor of Computer Science
Computer Science Department
University of California, Los Angeles
dt@cs.ucla.edu
3. **Masaki Nakada**
Co-Founder and Co-CEO
NeuralX Incorporated
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